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Sparse Identification of Nonlinear Dynamics-Based Feature Extraction for Data Driven Model Predictive Control of a Buck Boost Switch Mode Power Supply

This paper presents two methods of creating model predictive control (MPC) strategies for efficient real-time control algorithms for power electronics. Two novel methods of performing nonlinear modeling are presented in this research, the first being novel Takagi-Sugeno model, which combines two linear state space models using a membership function to model the nonlinear transitions between operating points. The second method involves using sparse identification of nonlinear dynamics (SINDy), a nonlinear modeling technique that uses $L1$ regularization of least squares for time-series data to define a parsimonious polynomial function set. This set is used to define the input feature space to extended dynamic mode decomposition (DMD) with control. These models are then used for data-driven model predictive control of a buck switch mode power supply to find the optimal duty cycle that regulates the output voltage over a finite tuned proportional integral derivative (PID) controller. Numerical accuracy challenges are discussed, and strategies are offered for their mitigation. [DOI: 10.1115/1.4065827]

1 Introduction

Switched mode power supplies (SMPSs) play an integral role in a variety of electronic devices due to their high efficiency and flexible power conversion by modulating the switching duty cycle of voltage-controlled switch. Their complex, nonlinear systems susceptible to several potential failure modes due to component degradation, thermal stress, electromagnetic interference, instability under varying load conditions, overcurrent and overvoltage situations, and potential design errors [1]. Over time, component characteristics can change leading to off-design operation, reduced efficiency, or catastrophic failure. High power levels, electromagnetic interference, and load variations can all lead to degradation that can suddenly destabilize the system if the control loop isn't robust enough [2]. This creates a growing need for robust, accurate, and adaptive control methods for SMPS to ensure reliable performance under varying operating conditions.

With the increase in computational speed and memory, one method that has become more prominent in the field of power electronics is model predictive control (MPC). The MPC method is an optimization problem that uses an explicit model of a dynamic system to predict future behavior over a finite receding horizon. The foundational work in MPC was presented in Refs. [3] and [4], which laid the groundwork for generalized predictive control for industrial processes. Work began to apply MPC methods to DC-DC power supplies in the early 2000s [5]. MPC continues to play a significant role in controlling SMPS devices, with recent research focused on

optimizing the implementation of MPC algorithms to further improve the speed and efficiency of SMPS, addressing the tradeoff between control performance and computational demand [6–8].

The MPC method requires adequate models to perform the receding horizon optimization problem and for SMPSs, the primary method is centered around state space average (SSA) models developed by Middlebrook et al. in the 1980s [9] and is still being used to simplify the nonlinear dynamics to create highly efficient control schemes [10]. The SSA method utilizes an infinite slew rate assumption to average the equivalent circuits for the device over each of their respective periods. If the current mode of operation presents a deviation in the equivalent circuit models, then the model would no longer represent the system well. Another solution for modeling power electronics are SPICE models, which use linear equivalent circuit models for a given circuit topology and can include increasing levels of realism, including self-heating effects [11].

The main issue with using these methods is that the SSA method does not model degradation effects of the switch and diode well and it is difficult to include SPICE quality models on on-board control logic. In light of the complexities of accurately modeling environmental and degradation impacts on SMPS, data-driven models integrated into MPC algorithms have come to the forefront [12]. Sparse identification of nonlinear dynamics (SINDy) algorithm, developed by Kaiser et al., provides a means to build nonlinear models from observational data for MPC problems [13]. Simultaneously, Mezic et al. used dynamic mode decomposition (DMD) to generate linear state-space models from fixed interval state data [14], an approach extended by Proctor et al. is able to account for input signals, termed as DMD with control and can reproduce the state and input matrix from a set of measurement states [15]. Further

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extending this approach to include an unlimited number of derived observables, referred to as extended DMD with control (eDMDc), provides a more flexible framework for high-dimensional data modeling [16].

This research outlines and evaluates two nonlinear, data-driven modeling strategies and their application in MPC for the improved modeling and control of switch-mode power supplies. The first strategy combines Takagi-Sugeno (TS) fuzzy models with logistic functions, adeptly navigating transitions between continuous and discontinuous conduction modes regulated by input voltage and load resistance. The selection of the TS fuzzy models was driven by the need to smoothly transition between continuous conduction mode (CCM) and discontinuous conduction mode (DCM) without abrupt changes. A logistic function was used to facilitate this gradual switch between modes. Regression analysis was employed to fit the logistic hyperparameters using LTSPICE, a powerful and widely-used SPICE-based analog circuit simulator developed by Analog Devices, simulations at different operating points, capturing the transition boundary between CCM and DCM and converting to the most representative model. The second approach harnesses the power of the SINDy algorithm and eDMDc. This combination identifies critical basis functions for a thorough portrayal of system dynamics, formulating a linear model for the MPC optimization problem. By utilizing SINDy to identify a parsimonious higher dimensional features space for the eDMDc, one can combine the power of nonlinear dynamic identification and high-dimensional linear state-space modeling, leading to more accurate, robust, and adaptable control solutions for SMPS systems.

Utilizing real-time system measurements, the proposed methodology provides robust fault tolerant control (FTC), which can adapt to changes in diverse operating conditions and ensuring efficient optimal performance. These two methods were chosen due to their unique ability to address specific challenges in modeling and controlling SMSP. The TS model effectively accounts for nonlinear shifts between CCM and DCM caused by changes in input voltage and load resistance. It blends two linear state-space models using a membership function, capturing nonlinear transitions between operating points, ensuring optimal MPC performance.

The primary contributions of this research lie in the development of MPC. Methodologies for a buck-boost SMPS using data-driven models are created from simulation. The paper introduces two innovative approaches for developing data-driven parametric models that are used to optimize the control strategy over finite receding horizons. The first approach involves a novel TS model, which blends two linear state-space models via a membership function to accurately capture the transition between continuous and discontinuous conduction mode operation. The second approach leverages SINDy, using L1 regularization to identify a parsimonious polynomial function set that represents the nonlinear dynamics of the system. This set defines the input feature space for eDMDc, leading to an efficient and accurate data-driven model. The eDMDc method, combined with SINDy, creates parsimonious parametric models that overcome limitations of Taylor and Padé series approximations. By using L1 regularization to select the most relevant features, eDMDc directly captures the nonlinear dynamics of the SMPS while avoiding the complexity and convergence issues of traditional approximation methods.

Both models are integrated into the MPC framework to find the optimal duty cycle that regulates the output voltage of the SMPS. Their performance is compared against a traditional proportional integral derivative (PID) controller, with simulations conducted using LTSPICE to demonstrate the superior accuracy and efficiency of the developed strategies. By accounting for parametric shifts in the underlying electrical attributes, the developed method has the potential to improve power management, for equipment that rely on SMPSs for precise voltage regulation through MPC using SINDy and eDMDc can reduce energy waste and maintenance costs. By using SINDy to identify a parsimonious polynomial model and eDMDc for accurate nonlinear approximation, the proposed approach handles varying load conditions with greater efficiency

and can improve energy efficiency and extend the lifetime of sensitive components by reducing output ripple and ensuring robust and reliable power supplies for uninterrupted service, potentially improving power supply performance across different sectors, such as telecommunications systems, renewable energy systems, and consumer electronics.

In the theory section, the innovative utilization of TS nonlinear modeling is revealed. This unique implementation proficiently manages the transitions in both continuous and discontinuous conduction modes by strategically incorporating logistic functions and considering factors like input voltage and load resistance. The foundational principles and mathematics behind the SINDy algorithm and the eDMDc method are elaborated in detail. These methods, instrumental in generating finite approximations of the critical Koopman operator, enable us to trace the dynamic spatial-temporal state transitions effectively. Following this, the application of MPC to these constructed models is presented. The focus lies in the intelligent design of constraints and cost functions, which ensures optimal control and robust modulation of duty cycles. In the result and discussion section, the report relies on simulation results to verify the effectiveness of the proposed data-driven methodologies. The section further addresses the existing challenges and envisages potential enhancements to improve the efficacy of the data-driven approach to MPC of power electronics. The report concludes by summarizing the key discoveries and insights drawn from the study.

2 Theory

2.1 Background on Switch Mode Power Supplies. Switched-mode power supplies are the cornerstone of modern power electronics, underpinning a wide array of electronic devices. A SMPS regulates the electrical power output using a switching regulator, effectively converting raw input power to the required level for the load. Among various topologies of SMPS, the buck converter, a DC-to-DC device, is a common choice for stepping down voltage efficiently. Figure 1 showcases the schematic of a buck converter, with equivalent circuit representations of all possible topologies given binary activation.

A buck converter operates in two stages. When the switch (usually a MOSFET) is on, the input voltage is applied across the inductor,

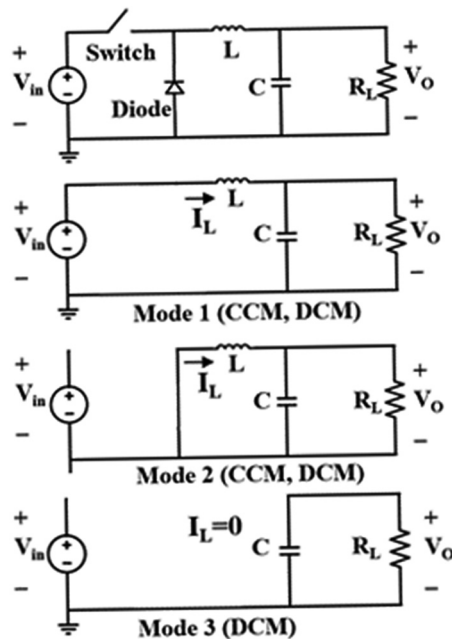


Fig. 1 Equivalent circuit diagrams of a buck SMPS. The three different possible “conduction modes” of the circuit are shown which depend on state of the MOSFET and diode state as well as the inductor current.

causing an increase in current while the diode is reverse-biased. When the MOSFET switches off, the stored energy in the inductor discharges to the load and the capacitor through the forward-biased diode. This dynamic process leads to a square wave of voltage across the inductor, resulting in a smoother current waveform due to the inductor's property of resisting changes in current. The capacitor refines the output voltage by smoothing the high frequency switching noise and providing a more constant DC output voltage.

The buck converter operates in two main modes: CCM and DCM. In CCM, the inductor current never falls to zero during the switching period, allowing for smaller inductor and capacitor sizes. DCM permits the inductor current to reach zero, enabling a zero current switching condition that reduces switching losses. The output capacitor absorbs the energy delivered by the inductor, and together they act as a low-pass filter to smooth out the square wave switching action of the MOSFET and diode into a stable DC output. For a Buck SMPS, for a given input voltage and load resistance, a specific critical inductance is needed that will cause the converter to operate on the boundary of the CCM and DCM operation, governed by Eq. (1) and presented in Fig. 2 [17]

$$L_C = \frac{R_L V_{in}^2}{2f_s (V_{in} - V_{out})^2} \quad (1)$$

2.2 State Space Averaging of Buck Switched Mode Power Supplies in Continuous Conduction Mode Operation. The SSA technique, pioneered by Dr. Middlebrook in the 1980s, is based on averaging the linear state space equations for each of the equivalent circuit models over their respective dwell period over the entire switching period. For the CCM operation, the inductor current stays above zero and the converter exists in either Mode 1 or Mode 2 equivalent circuit models.

For Mode 1, by performing nodal analysis and separating the differential terms, a linear state space can be formulated, seen below in the following equation:

$$\begin{aligned} \frac{dx(t)}{dt} &= A_1 \cdot x(t) + B_1 \cdot u(t) \\ \begin{pmatrix} \frac{di_L(t)}{dt} \\ \frac{dv_C(t)}{dt} \end{pmatrix} &= \begin{pmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{C \cdot R_L} \end{pmatrix} \cdot \begin{pmatrix} i_L(t) \\ v_C(t) \end{pmatrix} + \begin{pmatrix} \frac{1}{L} \\ 0 \end{pmatrix} \cdot V_{in} \end{aligned} \quad (2)$$

The same can be done for Mode 2 to form the linear state space seen below in the following equation:

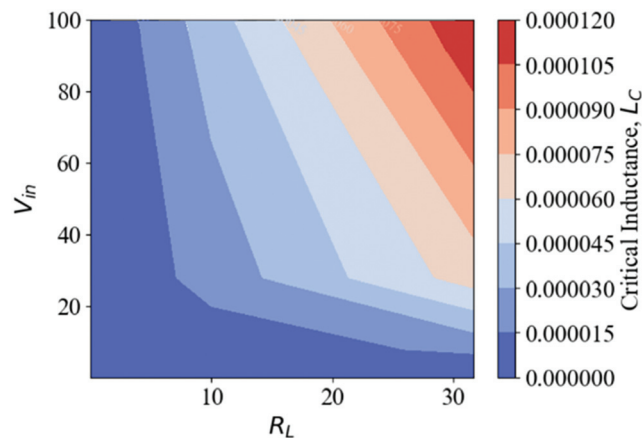


Fig. 2 Critical inductance values in terms of input voltage and purely resistive load. This value is the inductance which would place an ideal buck converter on the boundary of the DCM and CCM and the inductor current reaches zero directly in line with the switch period.

$$\begin{aligned} \frac{dx(t)}{dt} &= A_2 \cdot x(t) + B_2 \cdot u(t) \\ \begin{pmatrix} \frac{di_L(t)}{dt} \\ \frac{dv_C(t)}{dt} \end{pmatrix} &= \begin{pmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{C \cdot R_L} \end{pmatrix} \cdot \begin{pmatrix} i_L(t) \\ v_C(t) \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} \cdot V_{in} \end{aligned} \quad (3)$$

By averaging these models over the duty cycle, defined as $D = T_{on}/T_s$, the MOSFET “ON” period, T_{on} , over the switching period, T_s , an average linear state space can be formed that models the large signal aspects of the buck SMPS dynamics, seen below in the following equation:

$$\begin{aligned} \frac{dx(t)}{dt} &= (DA_1 + (1-D)A_2) \cdot x(t) + (DB_1 + (1-D)B_2) \cdot u(t) \\ \begin{pmatrix} \frac{di_L(t)}{dt} \\ \frac{dv_C(t)}{dt} \end{pmatrix} &= \begin{pmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{C \cdot R_L} \end{pmatrix} \cdot \begin{pmatrix} i_L(t) \\ v_C(t) \end{pmatrix} + \begin{pmatrix} \frac{1}{L}D \\ 0 \end{pmatrix} \cdot V_{in} \end{aligned} \quad (4)$$

2.3 Linearized Small Signal Model. The average model is a large signal model that can be used to model the transient nature of the converter, disregarding any higher harmonics from the switch and parasitic elements. To develop a small signal model, a Taylor series linear approximation between the duty cycle and state and any input variables is performed. This is done by setting the input voltage constant, $U = V_{in}$ setting the derivative to zero and solving for the average state and average duty cycle, $X = -A^{-1}B V_{in}$ and $D = V_C/V_{in}$. Linear perturbations about a mean signal are substituted for each state and input signal, and nonlinear terms are discarded, to form the following full-order small-signal model of the Buck SMPS used to make the time responses seen in results and discussion section:

$$\begin{aligned} \frac{d\tilde{x}(t)}{dt} &= A_{CCM} \cdot \tilde{x}(t) + B_{CCM} \cdot \tilde{u}(t) \\ \begin{pmatrix} \frac{d\tilde{i}_L(t)}{dt} \\ \frac{d\tilde{v}_C(t)}{dt} \end{pmatrix} &= \begin{pmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{C \cdot R_L} \end{pmatrix} \cdot \begin{pmatrix} \tilde{i}_L(t) \\ \tilde{v}_C(t) \end{pmatrix} + \begin{pmatrix} \frac{1}{L}D \\ \frac{1}{L}V_{in} \end{pmatrix} \cdot \begin{pmatrix} \tilde{v}_{in}(t) \\ \tilde{d}(t) \end{pmatrix} \end{aligned} \quad (5)$$

2.4 State Space Averaging of Buck Switched Mode Power Supplies in Discontinuous Conduction Mode Operation. In the DCM, the inductor current falls to zero during each switching cycle. Initially, when the switch is turned on, the inductor current rises from zero to a peak value. However, when the switch is turned off, the current in the inductor decreases, eventually reaching zero. At this point, no current is delivered to the load until the start of the next switching cycle. This zero-current period is important for achieving low losses and high efficiency in light load conditions.

For Mode 3, the same nodal analysis is performed as before and a linear state space is formed, seen below in the following equation:

$$\begin{aligned} \frac{dx(t)}{dt} &= A_3 \cdot x(t) + B_3 \cdot u(t) \\ \begin{pmatrix} \frac{di_L(t)}{dt} \\ \frac{dv_C(t)}{dt} \end{pmatrix} &= \begin{pmatrix} 0 & -\frac{1}{L} \\ 0 & -\frac{1}{C \cdot R_L} \end{pmatrix} \cdot \begin{pmatrix} i_L(t) \\ v_C(t) \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} \cdot V_{in} \end{aligned} \quad (6)$$

Following methods established in Ref. [4], a peak value of the inductor current is found using volt-second balance method:

$$i_{L,pk} = \frac{1}{L} * (V_{in} - V_C) * \frac{D_1}{f_s} \quad (7)$$

and used to define the DCM average model, seen in Eq. (9) below, in terms of D_1 and D_2 , the duty cycles in the first and second conduction mode, and

$$K_{DCM} = \begin{pmatrix} \frac{1}{D_1 + D_2} & 0 \\ 0 & 1 \end{pmatrix} \quad (8)$$

is a magnification term used to account for the third duty cycle in the state space equation

$$\begin{pmatrix} \frac{di_L(t)}{dt} \\ \frac{dv_C(t)}{dt} \end{pmatrix} = (D_1 A_1 + D_2 A_2 + (1 - D_1 - D_2) A_3) \cdot K_{DCM} \cdot \begin{pmatrix} i_L(t) \\ v_C(t) \end{pmatrix} + (D_1 B_1 + D_2 B_2 + (1 - D_1 - D_2) B_3) \cdot V_{in} \quad (9)$$

An operating point can be found, considering a constant input voltage, $U = V_{in}$ and using the assumption of zero capacitor in-series resistance, a gain for the circuit can be found using equation BCM, found in the following equation:

$$G = \frac{V_C}{V_{in}} = -\frac{R_L f_s}{4L} \pm \frac{1}{4L} \sqrt{R_L^2 f_s^2 + 8L R_L f_s} \quad (10)$$

The gain is then used to define the steady-state inductor current

$$I_L = G * \frac{V_{in}}{R_L} \quad (11)$$

and output filter capacitor voltage, $V_C = G * V_{in}$.

The Mode 2 duty cycle, D_2 , is found by setting the state derivatives in Eq. (9) equal to zero

$$D_2 = \frac{2L I_L}{\frac{D_1}{f_s} (V_{in} - V_C)} - D_1 \quad (12)$$

By performing the Taylor series linearization and rejecting the higher order terms, the small signal linear state space is found in Eq. (13) for the DCM condition and used to produce the waveforms.

$$\begin{aligned} \frac{d\tilde{x}(t)}{dt} &= A_{DCM} \cdot \tilde{x}(t) + B_{DCM} \cdot \tilde{u}(t) \\ \begin{pmatrix} \frac{d\tilde{i}_L(t)}{dt} \\ \frac{d\tilde{v}_C(t)}{dt} \end{pmatrix} &= \begin{pmatrix} \frac{2*V_C*f_s}{D*(V_C - V_{in})} & \frac{D^2*V_{in} + 2*I_L*L*f_s}{D*L*(V_C - V_{in})} \\ \frac{1}{C} & -\frac{1}{C*R_L} \end{pmatrix} \cdot \begin{pmatrix} \tilde{i}_L(t) \\ \tilde{v}_C(t) \end{pmatrix} \\ &+ \begin{pmatrix} \frac{D*(V_C - 2*V_{in})}{L*(V_C - V_{in})} & \frac{D^2*V_{in} + 2*I_L*L*f_s}{D*L*(V_C - V_{in})} \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} \tilde{v}_{in}(t) \\ \tilde{d}(t) \end{pmatrix} \end{aligned} \quad (13)$$

2.5 Buck Switched Mode Power Supplies Built in LTSPICE to Create Realistic Power Converter Data. An LTSPICE circuit used for simulation can be seen in Fig. 3. The simulation was used to model a buck circuit and compare the different modeling and control methodologies.

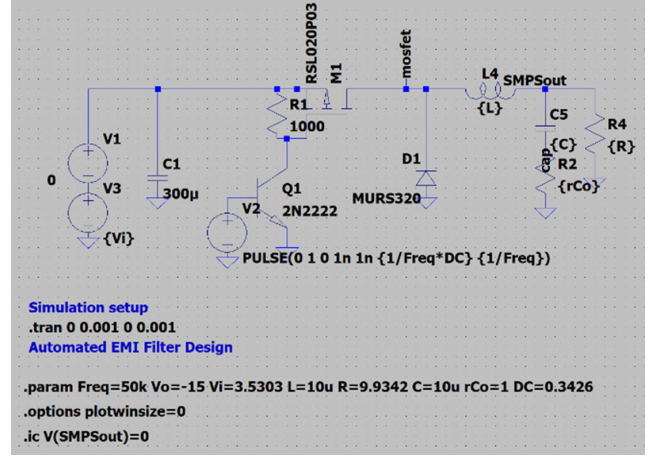


Fig. 3 Diagram of the LTSPICE model used to generate the training data for the data driven techniques. Piece-wise linear variable assignments were used to vary the input voltage, V_i , output load, R , and a behavioral voltage component was used to modulate the duty cycle, DC .

2.6 Data-Driven Method of Combining Continuous Conduction Mode and Discontinuous Conduction Mode State Space Average Models Using Takagi-Sugeno Method. Takagi-Sugeno (TS) fuzzy systems provide a data-driven modeling approach that integrates local linear subsystems according to the inputs' fuzzy sets. For a SMPS, a TS model can be created by assigning fuzzy sets to parameters such as load resistance and input voltage that blends the CCM and DCM linear models, leveraging a logistic function to gauge the proximity to the CCM-DCM boundary. This boundary is defined by the critical inductance, L_C , where the inductor current reaches zero at each switch cycle's end, as described by Eq. (1).

To build the TS model, a membership function

$$w = \frac{1}{(1 + \exp(-\alpha \Delta L))} \quad (14)$$

is used to govern which linear model is more dominant in the overall dynamic response, seen below in the following equation:

$$\frac{dx}{dt} = w(A_{CCM} \cdot x + B_{CCM} \cdot u) + (1 - w)(A_{DCM} \cdot x + B_{DCM} \cdot u) \quad (15)$$

The constant, α , ensures the exponent is between 0 and 1 and, $\Delta L = L_C - L$, where L is the converter inductance and L_C is the critical inductance found using equation BCM.

2.7 Koopman Operator Approximation Using Extended Dynamic Mode Decomposition With Control. The Koopman operator, K_d , is a linear operator that acts on an infinite set of basis functions to evolve a state in time within a dynamic process with perfect accuracy [15]

$$\begin{aligned} x[k+1] &= f(x[k]) \\ \Psi(x[k+1]) &= K_d \Psi(x[k]) \end{aligned} \quad (16)$$

The nonlinear basis functions lift the Koopman operator by moving from a finite-dimensional state space, made up of state variables, to a functional space, made up of potentially infinite observable functions of the state variables. Since the operator is linear, by truncating the basis functions, approximations of the Koopman operator can be acquired with incrementally decreasing error.

This method is called extended Dynamic Mode Decomposition (eDMD) and involves collecting extended basis features that represent snapshots of pairs of measurements that show how the dynamics evolve over time, $(x[k], x[k+1])$ for $k \in \{1, 2, \dots, j\}$, where k is the index of the observation taken over consistent time increments. The state variable, x , is a set of nonlinear real-valued functions, $\{\varphi_i : \mathbb{R}^n \rightarrow \mathbb{R}\}_{i=1}^N$, that make up the set of observables, $\varphi_i(x) = x_i, i = 1, 2, \dots, n$.

These extended features are formed into two matrices X_1 and X_2 to represent the dynamic systems evolutionary relationships

$$X_1 = \begin{pmatrix} \vdots & \vdots & \vdots \\ x[1] & x[2] & \cdots & x[j-1] \\ \vdots & \vdots & \vdots \end{pmatrix} \quad (17)$$

$$X_2 = \begin{pmatrix} \vdots & \vdots & \vdots \\ x[2] & x[3] & \cdots & x[j] \\ \vdots & \vdots & \vdots \end{pmatrix}$$

In the eDMDc method, the problem of solving for the A and B matrices in $X_2 = AX_1 + BU$ is cast as a least squares problem

$$X_2 = \begin{pmatrix} X_1 \\ U \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} \quad (18)$$

Singular value decomposition is used to produce a diagonal matrix of singular values and left- and right-hand eigenvector matrices. By choosing the first, r columns of which, the noise and computational complexity can be truncated. With

$$Z = \begin{pmatrix} X_1 \\ U \end{pmatrix} \quad (19)$$

$$Z = W \Sigma V^H$$

and the reduced order projections of the A and B matrices are found from the following least squares problem:

$$X_2' W' = V \Sigma \begin{pmatrix} A_p \\ B_p \end{pmatrix} \quad (20)$$

Full order state and input matrices are then reconstructed, below in Eq. (21), using the left-hand eigenvector from the singular value decomposition, but this is not necessary if the point of the exercise is to come up with a reduced order model to save on computational complexity of the MPC algorithm.

$$\begin{pmatrix} \tilde{A} \\ \tilde{B} \end{pmatrix} = \begin{pmatrix} U \cdot A_p \cdot U^H \\ U \cdot B_p \end{pmatrix} \quad (21)$$

2.8 Sparse Identification of Nonlinear Dynamics. The SINDy algorithm is a data-driven approach for discovering the governing equations of a dynamical system from measurement data. It operates by constructing a library of candidate basis functions, applying sparse regression to identify the most significant features, thereby constructing a parsimonious model of the system. For a dynamical system represented by Eq. (22), SINDy seeks to find a sparse representation of the function, f , by solving the following sparse regression problem:

$$X_2 = \Xi \Theta(X_1) \quad (22)$$

where Θ is a basis function library that extends the state and input features by including all combinations of second-order polynomial expansions of the vector x_m , denoted $X_m^{P_2}$, for the state and input features

$$\Theta = \begin{pmatrix} \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & X_1 & X_1^{P_2} & \cdots & U & U^{P_2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix} \quad (23)$$

The sparse matrix, $\Xi = (\xi_0 \ \xi_1 \ \cdots \ \xi_n)$, is made up of the predictor variables that dictate which functions have the most influence in the overall dynamics and are found using the LASSO regression algorithm [3]

$$\min_{\xi_k} \|X_{2,k} - \xi_k \Theta(X_1)\|_2^2 + \lambda \|\xi_k\|_1 \quad (24)$$

where λ is the sparsity constraint of the predictor variables of the k -th row of the predictor variables, ξ_k . The terms identified by Ξ are used to truncate the infinite basis functions, $\Psi(x_m)$, which act on the Koopman operator to approximate the dynamics within tolerable accuracy.

2.9 Proportional Integral Derivative Control. To compare the different model-based control techniques, a proportional integral derivative (PID) control is proposed. This method is commonly used in SMPS devices because it can be implemented in a simple double zero Op-Amp configuration that is excited by a simple timer circuit. This method is agnostic to the underlying state of the circuit and super imposes three duty cycle corrections based on the error of the signal. The PID control logic can be found below in Eq. (25), in terms of the PID gains, K_p , K_i , and K_d .

$$\Delta d = K_p * (V_{out} - V_{ref}) + K_i * \int (V_{out} - V_{ref}) dt + K_d * \frac{d}{dt} (V_{out}) \quad (25)$$

2.10 Model Predictive Control. Model predictive control is a strategy that uses a model of the system to predict future outputs. The research applies the MPC method to optimize the state output response along with the duty cycle control action to minimize a cost function over a defined horizon, given the box constraint on the duty cycle ($0 \leq d \leq 1$). The objective of the controller can be seen in Eq. (26) below, which seeks to minimize the deviation of the output voltage from a reference voltage, and the sum of squares of the duty cycle for effective output voltage regulation and state and input response reduction

$$\min_{\hat{u}_d(k)} J(\hat{u}_d(k)) = \min_{\hat{u}_d(k)} \left\{ \sum_{i=1}^N (\hat{v}_O(k+i+1) - V_{ref})^2 + \lambda \hat{u}_d(k+i) \right\}$$

Subject to

$$\begin{aligned} \hat{x}(k+1) &= A \cdot \hat{x}(k) + B \cdot \hat{u}(k) \\ 0 &\leq \hat{u}_d(k) \leq 1 \end{aligned} \quad (26)$$

The $\hat{u}_d(k)$ in the constraints signifies a sequence of duty cycle control inputs over the prediction horizon of N time increments, where $k \in \{0, 1, \dots, N-1\}$, V_{ref} is the desired reference voltage, and λ is a weighting factor that trades off on tracking performance and control effort.

2.11 Performance Calculations. To evaluate and compare the performance of the different control scheme, the root-mean-square error (RMSE), found in Eq. (27), and the converter efficiency, defined as the output power over the input power, found in Eq. (28), were used

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2} \quad (27)$$

$$\eta = \frac{P_{out}}{P_{in}} = \frac{V_O^2}{R} \frac{1}{V_{in} I_{in}} \quad (28)$$

3 Results and Discussion

The traditional modeling techniques of state space averaging do not address the bifurcation between CCM and DCM in state space. This research combined state-space models for both conditions using the TS framework and evaluated them with RSME over the input voltage and load resistance range. The LTSPICE simulation was run at multiple operating conditions using a Latin hypercube sampling routine to sample the input voltage, V_{in} , load resistance, R_L , and duty cycle, d , using piecewise linear look-up tables programmed into the transient simulation response. The data were used as training data input for the SINDy, eDMDc system identification routines, and identifying the shape parameter, α , in the TS method. The results of the findings are presented below.

3.1 Sparse Identification of Nonlinear Dynamics Parametric Model Results. To gain interpretability into the model and understand which features are most dominant and should be included in the extended features of observables, the SINDy approach was used, outlined in theory section. To evaluate the different extended features in the polynomial expansion of the state and input data, the scaling of the different relationships had to be standardized. Normal noise of zero mean and one standard deviation was added to each signal collected from the output response of the LTSPICE program and the data was standardized and used as features into the 100 iterations of the SINDy program. The coefficients value distributions were collected and assembled into Fig. 4.

The coefficients and features can be assembled into the following nonlinear state space model used to make the time responses seen in Figs. 5 and 6:

$$\begin{aligned} \frac{dv_O}{dt} = & -313.342 v_O - 7452.3 i_L + 660.1 v_O^2 \\ & - 461.1 v_O i_L - 52.1 i_L^2 + 487.2 v_i \\ & - 75.9 r_L - 821.4 d - 452.7 v_i^2 + 26.0 v_i r_L \\ & - 100.1 v_i d - 49.7 r_L^2 \\ & + 20.2 r_L d - 109.7 d^2 \\ \frac{di_L}{dt} = & 46091.770 v_O - 11119.6 x_1 - 15868.0 v_O^2 \\ & - 6800.6 v_O i_L - 267.5 i_L^2 \\ & - 46007.5 v_i + 1160.6 r_L - 4945.5 d \\ & + 12171.1 v_i^2 - 368.3 v_i r_L \\ & + 5736.6 v_i d + 938.7 r_L^2 - 339.5 r_L d + 4441.4 d^2 \end{aligned} \quad (29)$$

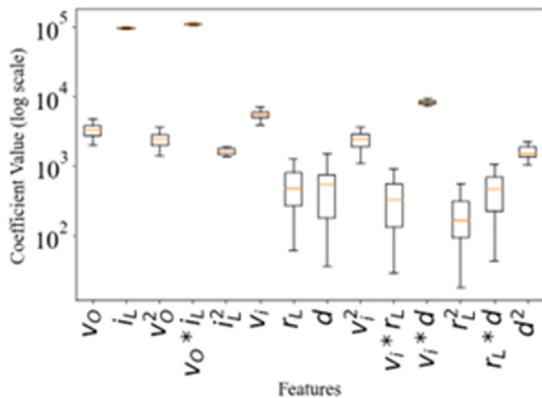


Fig. 4 Coefficient values for the standardized extended features. Values were generated by including noise with the LTSPICE training data and standardizing the signals to use as inputs into the SINDy algorithm. Standardizations allow for determinations to be made for which terms are most dominant in the underlying dynamics.

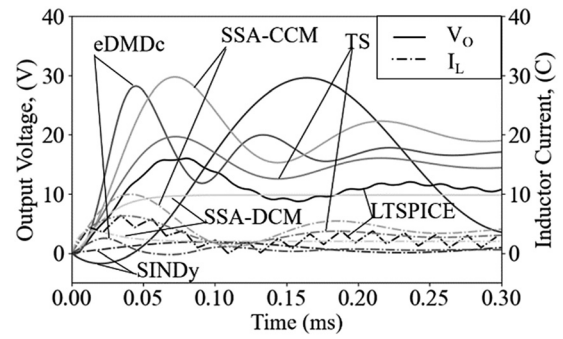


Fig. 5 Time-series response for the different CCM SSA models, TS model created using the critical inductance relationship, the nonlinear model derived from the SINDy method, the eDMDc method using the truncated extended features to solve the data-driven controls problem, and the LTSPICE model shown in Fig. 3, with moderate voltage ripple and large amount of inductor ripple current

3.2 Extended Dynamic Mode Decomposition With Control Model Results. The extended states verified in the SINDy program are used to extend the features space past the state variables. The states used in this analysis are made up of a polynomial of the output voltage, v_O and the inductor current, i_L . The inputs used are made up of the polynomials of the input voltage, v_i , load resistance, r_L , and duty cycle, d

$$\begin{aligned} x &:= \{v_O, i_L, v_O^2, v_O i_L, i_L^2\} \\ u &:= \{v_i, r_L, d, v_i^2, v_i r_L, v_i d, r_L^2, r_L d\} \\ i &\in \{1, \dots, N\} \end{aligned} \quad (30)$$

The quadratic feature, d^2 , presents the most difficulty in terms of solving the constrained optimization problem, and from Fig. 4, the coefficient of the standardized features has a low weighting value. By discarding the d^2 -parameter, along with any state-input interactions for inclusion into the extended features used to perform the eDMDc, the MPC optimization problem remains convex, and will not have a problem with local minima.

The numerical results of the full order transformation in Eq. (21) can be found below. These matrices were used, along with the extended state and input features, to create the time series plots seen in Figs. 5 and 6.

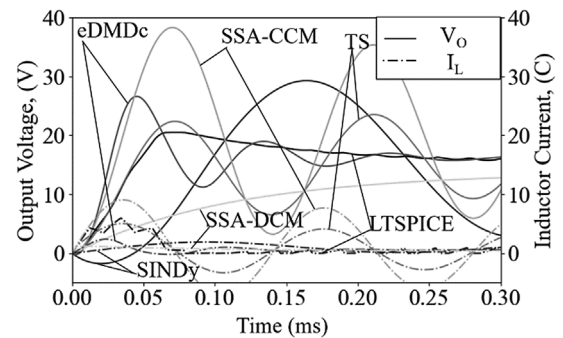


Fig. 6 Open loop time-series response, using parameters in Table 1, for the DCM SSA models, TS model created using the critical inductance relationship, the nonlinear model derived from the SINDy method, the eDMDc method using the truncated extended features to solve the data-driven controls problem, and the LTSPICE model shown in Fig. 3

$$\tilde{A} = \begin{pmatrix} 0.9891 & -0.5282 & 0.0001 & -0.0025 & -0.0111 \\ 0.007 & 1.0156 & 0 & -0.0012 & -0.0023 \\ -0.3455 & 0.6171 & 0.9949 & -1.3064 & -0.8032 \\ -0.204 & 3.9686 & 0.0059 & 0.8481 & -0.605 \\ 0.0462 & -1.8858 & -0.0005 & 0.0515 & 1.0427 \end{pmatrix}$$

$$\tilde{B} = \begin{pmatrix} 0.0051 & -0.0022 & -0.3882 & 0 & -0.0021 & 0.0024 & 0 & 0 \\ -0.0022 & 0.0004 & -0.0751 & 0 & -0.0028 & -0.0002 & 0 & 0 \\ -0.0638 & 0.0198 & 10.8896 & 0.0007 & -0.8441 & 0.0062 & 0.0039 & -0.0005 \\ 0.3299 & -0.0585 & 2.9289 & 0 & -0.3001 & 0.0725 & -0.0047 & 0.0002 \\ -0.0494 & 0.0077 & -2.0967 & 0 & 0.058 & -0.0081 & 0.0004 & 0 \end{pmatrix}$$

3.3 Model Time-Series Results. To compare the different models to the actual time-series data generated from LTSPICE, time-series plots were generated for a buck SMPS in CCM and DCM operation, found in Figs. 5 and 6, respectively, using parameters for the models found below in Table 1.

The time response for the CCM condition for the LTSPICE circuit and the state space, SINDy, and eMDMc models is found in Fig. 5, and shows the TS model doing well at capturing the output voltage. The data-driven models are trained on limited averaged data and will not show the ripple characteristic of the LTSPICE behavior and may be less likely to fit specific operating points in favor of generalizing the global behavior. The SINDy and eMDMc models are also limited in that were restricted from using terms that reflect state-input interaction which may explain why they did not perform well compared to the TS model.

For the time response modeling of the DCM case, the eMDMc method did the best at modeling the output voltage, with exception of not all the transient characteristics being accurately portrayed, which is essential to accurately predicting the phase response. The shape parameter for the TS method was found qualitatively to be around $\alpha = 0.3$. The eMDMc, TS, and DCM SSA method models the transition from CCM to DCM by accurately modeling the inductor current phenomena in both the CCM and DCM case, which is something the individual DCM CCM models built from the analytical state space fail to do.

Overall, it seems that both the eMDMc and TS models would make good candidates for applying the constrained MPC method to the SMPSs.

3.4 Control System Verification Comparison of Tracking Error Over Time. To compare the two data-driven modeling MPC approach with another recognized efficient method of closed-loop feedback control, a PID controller was formulated to make duty cycle corrections. The PID gains were tuned at an operating point of $V_{in} = 20$ V and $R_L = 20 \Omega$ and were found to be $K_P = 0.01$, $K_I = 0.001$, $K_D = 0.01$.

The λ -value associated with Eq. (19) of the MPC algorithm was found to be $\lambda = 10^{-30}$ for the eMDMc method and $\lambda = 10$ for the TS

method. The eMDMc and the PID methods are both comparable at the operating point, with the eMDMc method having slightly better response, where-as the TS had much larger overshoots in the corrections and seemed to be reacting to the noise of the converter caused by the switching, and nonideal input filter capacitor and output filter capacitor (Fig. 7).

3.5 Comparison of Root-Mean-Square Error and Efficiency, η , Over Different Operating Conditions. The developed models were specifically designed to handle a broad range of parameter variations, as demonstrated by evaluating the RMSE and efficiency across a wide operating range of 0–95 ohms load resistance and 0–100 volts input voltage. During data simulation, a 10% white noise signal was added to the input voltage to test robustness against noise. Smoothing functions were applied to the data to mitigate signal noise, ensuring that the models could maintain accurate performance even under noisy conditions. To compare the two data-driven modeling methods, TS and the eMDMc method both were used in the MPC algorithm, and the response control signals were used as input into incremental LTSPICE simulations. The RMSE was evaluated using Eq. (20), and heat maps of the results were prepared in Fig. 8. Overall, the MPC and PID method were able to compensate the output voltage with a small amount of error over a similar range of output voltages and load resistances and tend to go unstable within the same regions. Loads above $R_L = 70 \Omega$ cause the eMDMc tracked output voltage response suddenly go to zero. The TS method only shows low error around one load setting for the input voltage settings.

The efficiency was also evaluated using Eq. (21), and heat maps were generated from the results prepared in Fig. 9. Overall, the

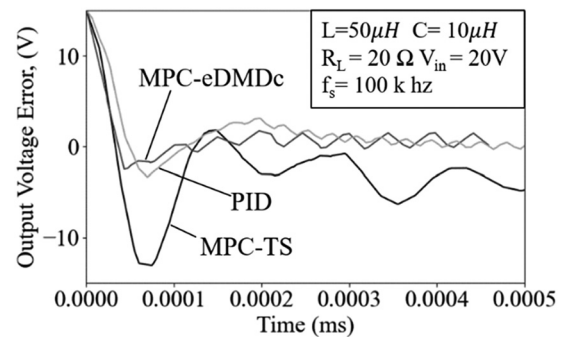


Fig. 7 Closed loop time-series response of the output voltage error for the DCM SSA models, TS model and the PID compensation method applied to incremental LTSPICE simulations using the circuit found in Fig. 3

Table 1 Buck boost SMPS parameters used in the simulation and state space models

Nomenclature	CCM value (Unit)	DCM value (Unit)
Inductance, L	50 μ H	50 μ H
Capacitance, C	10 μ F	10 μ F
Load resistance, R_L	40 Ω	5 Ω
Input voltage, V_i	20 V	20 V
Frequency, f	100 kHz	100 kHz
Duty cycle, D	0.5	0.5

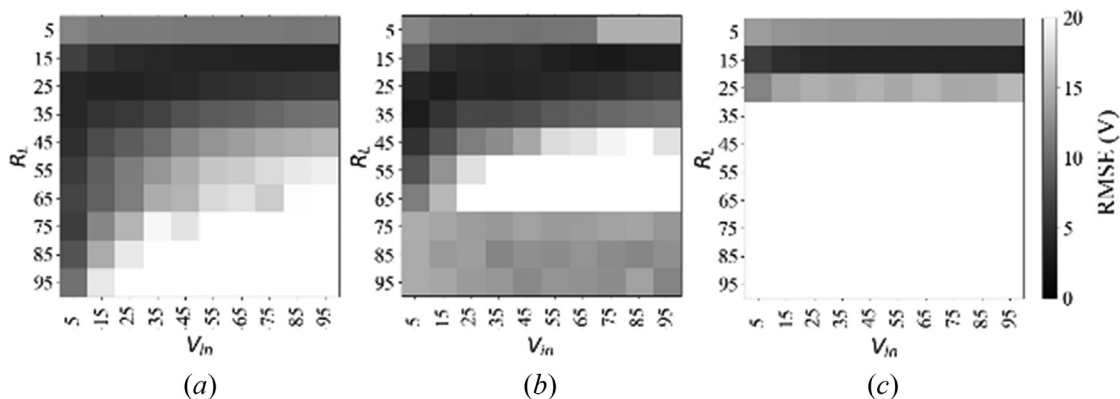


Fig. 8 Heat maps of RMSE for the three closed-loop compensation methods applied to the LTSPICE simulation: (a) PID, (b) MPC-eDMDC, and (c) MPC-Takagi-Sugeno

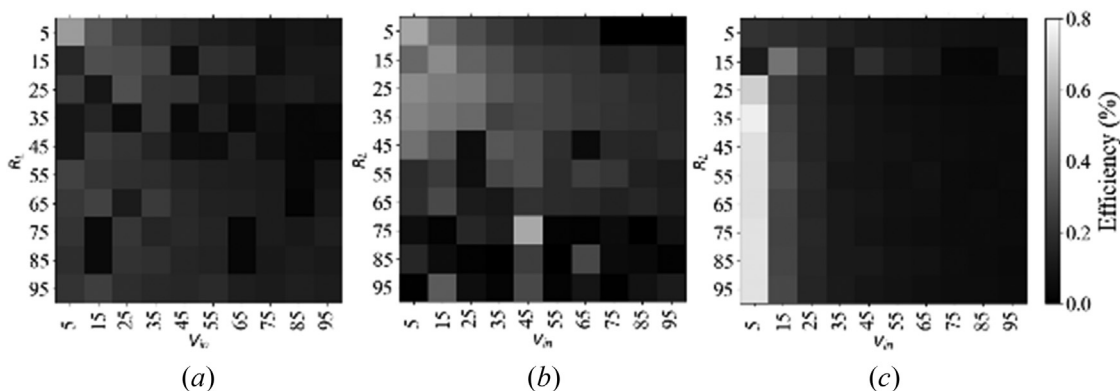


Fig. 9 Heat maps of efficiency, η , for the three closed-loop compensation methods applied to the LTSPICE simulation: (a) PID, (b) MPC-eDMDC, and (c) MPC-Takagi-Sugeno

MPC-eDMDC method shows more efficiency over larger region of input conditions compared to the PID compensator. The MPC-TS method showed high efficiency at only one point in the envelope that overlapped low RSME error, which was at the $R_L = 20 \Omega$, $V_{in} = 20 \text{ V}$.

3.6 Discussion. The results show that the MPC-eDMDC method can be used to create a robust compensation scheme for a buck SMPS, given there is enough data to model the necessary operating points. The TS method requires less data in that it is built from existing domain knowledge concerning the detail of the SSA models, more work is required to map the membership functions to account for the DCM to CCM transition.

Sparse identification of nonlinear dynamics model results were limited to the fact that the state-input interactions were discarded to be able to form the augmented least squares optimization problem required for the control problem, seen in Eq. (20). More work needs to be done to include more times of features into the state embedding. Time delay embedding and state derivative embedding offer good possibilities to maximize the impact of the SINDy feature selection routine by increasing the dimensionality and including spatiotemporal information to further extend the state features.

The MPC-eDMDC method performed well at improving the LTSPICE model time-response. While not having a high efficiency compared to current benchmarks [10], the purpose of this research was to apply novel data-driven approaches to improve upon existing SMSP devices, without the need to perform laboratory testing to determine the hidden electrical attributes. With a lower RMSE and higher efficiency in certain areas of the envelope, the MPC-eDMDC method does outperform the PID controller, but the PID controller is able to regulate a larger swath of input states and more work needs to be done to improve the model predictive approach at these points. In terms of accuracy, the MPC method outperformed the traditional PID controller at lower load ranges but declined at higher loads. The

TS model worked within a narrow load resistance range, while the PID controller had a broad accuracy range across 0–100 ohms load resistance and 0–100 volts input voltage. The SINDy algorithm, combined with eDMDC, provided accurate nonlinear modeling without relying on specific operating points. The eDMDC-MPC method of controlling the buck boost SMPS demonstrated slightly higher computational efficiency than the PID controller for most parts in the operating envelope.

3.7 Contributions. The main contribution of this research is a novel approach to data-driven modeling of SMPSs produced by employing SINDy eDMDC for feature extraction and model building. This technique demonstrates the potential for capturing the nonlinear dynamics of SMPS, particularly being able to model the transition between CCM and DCM operation and model the complex influence of internal parasitic elements without complicated analytics, which is a significant advance in developing methods to account for parameter shift degradation problem with minimal changes in the circuit topology and only using output voltage and inductor current as input variables.

The method that was developed performs optimal and robust duty cycle modulation in real-time, based on the SINDy, eDMDC, and TS predictive models using historical data. This allows for proactive management of system behavior based on predicted parameter shifts to achieve a robust FTC. This demonstration of a novel methodology for FTC in SMPS is a significant achievement that can greatly improve system reliability, and lead to greater energy efficiency and stability.

3.8 Recommendations for Future Work, Improvements, and Potential Applications. One main area of interest in improving upon this specific case study would be to use a functional approach to collecting the training data. This would help surpass the low-data

limit necessary for the data driven methods to work by allowing for relaxation of sampling schemes.

Deep learning methods present a lot of benefit in terms of creating highly accurate models, but these models are opaque in how they make decisions. It has been shown that reducing the size of a neural network to a level comprehensible to humans by handcrafting input features to provide more interpretable models while retaining the same predictive power [18]. This route can also be explored to define the predictive models used in the MPC routine.

To improve the TS routine, more detailed CCM and DCM models can be created by including parasitic impedances. Another approach that could improve the MPC-TS method is to perform a practical assessment of critical inductance for a given SMSP topology at the DCM CCM boundary instead of relying on Eq. (1), derived for an ideal converter. These additional considerations would lead to a more accurate model, and better predicted feedback compensation when applied to the MPC method.

4 Conclusion

In this study, we have taken an innovative approach to optimize SMPSs by utilizing data-driven methods, particularly eDMDc and SINDy, in conjunction with MPC. Our approach has the unique advantage of capturing and modeling the complex nonlinear dynamics of SMPS, particularly at the transition between CCM and DCM, which is often a challenge in conventional methods. By doing so, we have significantly improved the potential for robust FTC, leading to improved system reliability, enhanced stability, and superior energy efficiency. Despite the challenges associated with data intensiveness and the need to refine the SINDy and TS models, our research represents a substantial stride forward in SMPS modeling. Our findings suggest promising pathways for future work, which include refining data collection techniques, improving model precision, and expanding the capability of the MPC approach.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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